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A paleomagnetic and rock magnetic study of Tertiary volcanics from the Vogelsberg (Germany)

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ABSTRACT

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Paleomagnetic results are presented for orientated cores from 37 sites in the Vogelsberg (Germany) Tertiary volcanic province. The available radiometric dates indicate that the main volcanic activity occurred between 18 and 15.5 Ma (Early Miocene) with intermittent, mainly intrusive, activity continuing into the Late Miocene. A stable natural remanent magnetization (NRM) direction was obtained from most sites by stepwise alternating field (AF) demagnetization. The mean direction (excluding sites with intermediate NRM directions) is: $D = 358.6°$, $I = 64.0°$, $\alpha_{95} = 5.2°$, $N = 31$. This gives a virtual geomagnetic pole (VGP) of 85.1° N, 200.9° E, which agrees well with the Late Tertiary apparent polar wander path (APWP) for Eurasia. A reverse–normal–reverse magnetostratigraphy is observed, which can be correlated with chrons 5Cr, 5C, and 5Br (17.6–15.2 Ma).

Results of rock magnetic and reflected light microscopy studies indicate that titanomagnetite and its oxidation products are the main magnetic phases present. The grain size and degree of deuteric oxidation varies considerably between sites, with the titanomagnetites of thin vesicular lava flows tending to be more oxidized than those in thick massive bodies. Low-temperature susceptibility measurements show a range of behaviour, with some samples possessing grains which unblock below room temperature.

1. Introduction

The Vogelsberg forms part of a 700-km-long belt of Tertiary volcanism which stretches across central Europe from the Eifel in the west to Lower Silesia in the east (Lippolt, 1982). It is the largest extinct volcano in central Europe, with a present diameter of 60 km at its widest point (Ehrenberg, 1986). Paleomagnetic measurements were made in the 1950s by Angenheister (1956) and Nairn (1960). Since that time, paleomagnetic studies have been limited to the measurement of the inclination of the NRM and susceptibility from borehole samples (Angenheister and Turkowsky, 1964; Schenk, 1968; Harre et al., 1975; Ehrenberg et al., 1981). Since the last pole positions were published for the Vogelsberg, there have been significant advances in paleomagnetic techniques and equipment in particular, the use of stepwise alternating field (AF) demagnetization. Of the central European volcanics studied by paleomagnetists (Weinreich and Bleil, 1984), the Vogelsberg is the only area without demagnetized NRM data. As part of a project to look at how variations in rock magnetic characteristics affect paleointensity determinations, the paleomagnetic and rock magnetic properties of a new collection of orientated cores from the Vogelsberg have been studied.

2. Geology

The Vogelsberg volcanic massif lies just to the north of the Rhine Graben, but is not directly

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related to it (Ehrenberg, 1986). Analysis of borehole data (Ehrenberg and Hickethier, 1985) reveals that the basalt base of the Vogelsberg has a block-like structure, with a NE–SW trending zone of greatest subsidence, which was already active before the onset of volcanic activity. The eroded volcanics today cover an area of some 2000 km^2, with a maximum thickness in excess of 490 m (Ehrenberg et al., 1981).

The oldest volcanics have been found in boreholes in the central Vogelsberg (Ehrenberg et al., 1981). These are an irregular sequence of alkali basalts and tuffs as well as more differentiated alkali rocks (including trachytes). In younger sequences, tholeiitic basalts are found in addition to the alkali rocks. Attempts to define a stratigraphy are sometimes complicated by shallow intrusions of alkalic magma; for example, at Vogelkopf quarry (site 4A/4B) (Ehrenberg et al., 1977; Ehrenberg, 1986), where the lower basalt in the sequence of basalt/brown coal/basalt is younger than the overlying coal.

TABLE 1

Summary of the eruptive history of the Vogelsberg using the available K–Ar dates (recalculated using new decay constants (Steiger and Jäger, 1977) from Ehrenberg et al. (1977, 1981), Harre et al. (1975) and Kreuzer et al. (1973, 1974)

High Vogelsberg (borehole data)	
> 400 m of alkali olivine basalts, trachytes and tuffs: mean age 17.6 ± 0.2 Ma	
SW Vogelsberg (Rainrod I borehole)	
alkali olivine basalts and tholeiitic basalts: 17.5 ± 0.2 Ma–16.6 ± 0.2 Ma	
with some younger (13.3 Ma) (?) intrusives (or Ar loss)	
NW Vogelsberg (Londorf area)	
alkali olivine basalts and tholeiites:	
mean age	16.1 ± 0.3 Ma
range	17.0–15.5 Ma
with later intrusions: 12–10 Ma	
South Vogelsberg (Vogelkopf)	
two flows: lower	16.3 ± 0.3 Ma
upper	14.9 ± 0.5 Ma
extrusive and subfusive alkali olivine basalts	
Main activity 17.8–15.5 Ma, extrusive and shallow intrusive	
Intrusive activity continued intermittently until ~10 Ma	

In general, the oldest rocks are found in the centre of the present-day outcrop, where the erosion levels are deepest.

The warm climate during the Upper Miocene provided suitable conditions for the deep weathering of the volcanics to form hematite-rich deposits (basalt–ironstone), red clay and bauxite (Ehrenberg, 1986). These are found today predominantly in the western Vogelsberg — this is presumably a reflection of differential erosion.

K–Ar data (summarized in Table 1) (Kreuzer et al.,1973, 1974; Harre et al., 1975; Ehrenberg et al., 1977, 1981) indicate that volcanic activity was at its peak in the Burgidalian (18–15.5 Ma), but continued intermittently for another 5 Ma, although it seems that some of the younger K–Ar dates may be caused by loss of radiogenic argon (Ernst, 1977; Ehrenberg et al., 1981).

3. Paleomagnetism

Between four and 10 orientated 2.5-cm drill cores were taken from a total of 37 sites in quarries or roadcuts distributed throughout the Vogelsberg (Fig. 1, Table 2). Unorientated hand-samples, for rock magnetic and paleointensity studies, were taken from four additional sites.

Three or more samples from each site were measured using a Digico and/or Minispin spinner magnetometer and demagnetized by alternating field (AF) in 5-mT steps to 50 mT, and then in 10-mT steps to 90 mT. Examples of demagnetization behaviour are shown in Fig. 2. The direction of the stable magnetization was determined for each sample by displaying the data on a Zijderveld (1967) orthogonal plot and using a computer program (Sherwood, 1989) to fit a line by principal component analysis (Kirschvink, 1980) to the linear segment representing the primary remanance direction.

The results show that the samples have, in general, a single-component NRM, which is often partially obscured by a viscous component. As part of the paleointensity experiments, thermal demagnetization data were obtained for some samples (Fig. 3). The results confirm the single-component nature of the magnetization.

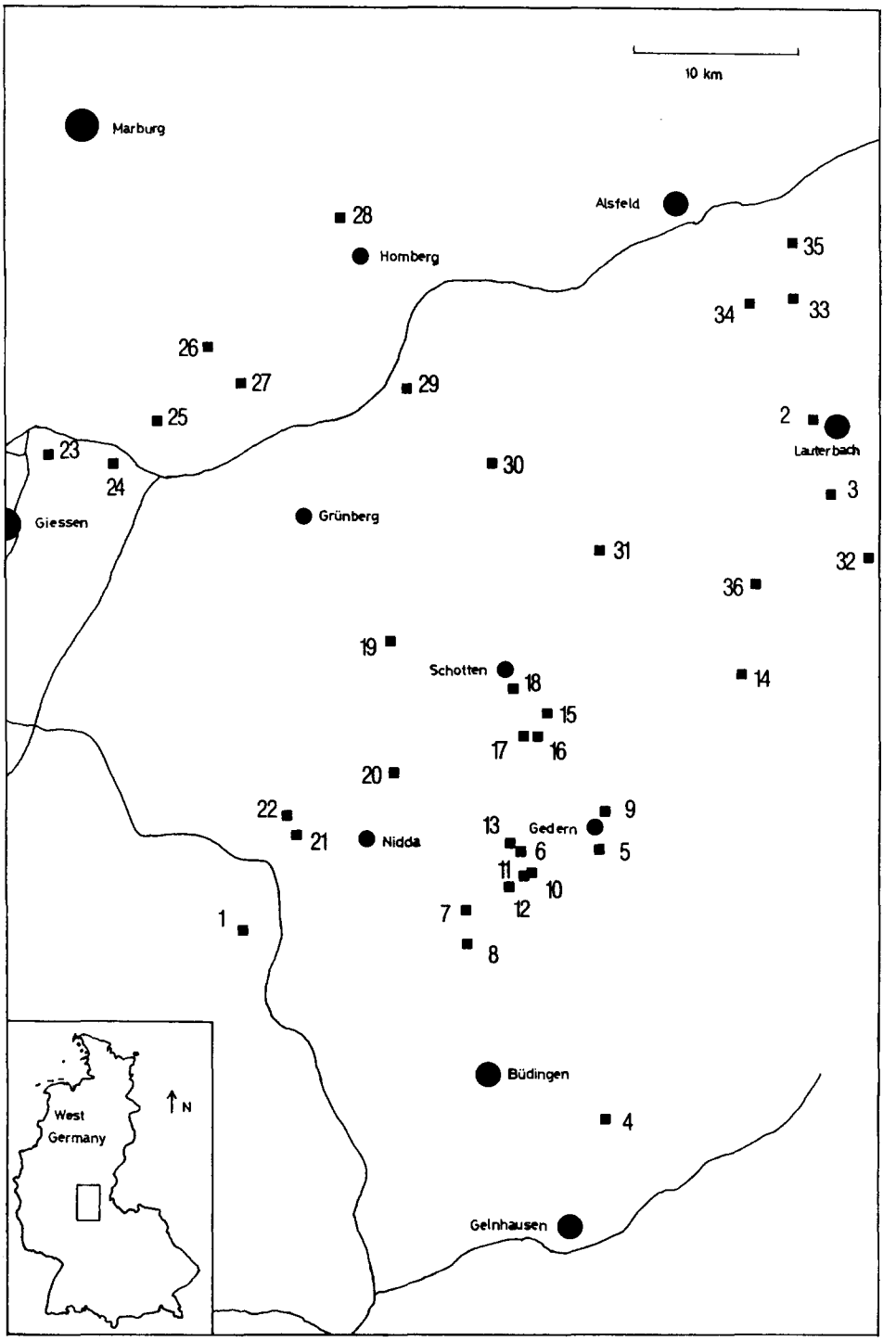


Fig. 1. Map showing the sampled localities.

TABLE 2

Details of sample sites and paleomagnetic results

Site	Locality	Position (R, H)	n/N	D	I	α_{95}	Polarity
1	Bingenheim	359300,558110	–	–	–	–	–
2A	Rimlos	352690,561190	3/3	344.2	50.3	3.6	N
2B	Rimlos	352690,561190	3/3	356.8	56.4	2.2	N
2C	Rimlos	352690,561190	3/3	351.9	52.5	6.6	N
3	Eisenbach	352780,560730	3/3	110.3	−65.3	7.3	I
4A	Breitenborn	351485,556990	3/3	186.7	−60.5	4.0	R
4B	Breitenborn	351485,556990	3/3	67.7	43.0	11.2	I
5	Gedern	351415,558640	3/4	209.2	−53.8	10.2	R
6	Glashütten	350960,558595	3/3	119.4	−70.0	2.6	R
7	Lissberg	350650,558260	–	–	–	–	–
8	Usenborn	350670,558035	3/3	188.6	−60.7	11.8	R
9	Gedern	351425,558840	3/3	190.4	−24.0	7.0	R
10	Hirzenhaim	351000,558455	2/4	208.1	−75.3	7.2	R
11	Hirzenhaim	351020,558465	3/3	171.0	−76.9	8.4	R
12	Hirzenhaim	350930,557385	3/3	143.8	−51.8	3.7	R
13	Glashütten	350910,558650	3/3	135.8	−69.0	5.2	R
14A	Grebenhain	352300,559660	3/3	242.4	−42.8	19.6	I
14B	Grebenhain	352300,559660	3/3	346.7	68.7	7.2	N
15	Eschenrod	351130,559390	3/3	190.5	−76.0	9.5	R
16	Eschenrod	351065,559305	3/4	200.7	−80.1	3.8	R
17	Wingershausen	350980,559270	3/3	124.6	−63.9	5.3	R
18	Schotten	350900,559570	3/3	173.1	−74.1	4.2	R
19	Gonterskirchen	350180,559855	3/3	225.3	−65.1	7.4	R
20	Oberschmitten	350220,559065	3/3	171.2	−58.2	4.3	R
21	Häuserhof	349630,558685	0/3	–	–	–	I
22	Oberwiddersheim	349590,558800	3/3	168.6	−49.7	11.6	R
23	Alten Buseck	338140,560935	3/3	186.4	−20.1	6.4	I
24	Grosse Buseck	348540,560870	–	–	–	–	–
25	Beuern	348792,561168	3/3	152.8	−60.1	4.9	R
26	Londorf	349105,561600	3/3	33.2	70.3	1.5	N
27A	Odenhausen	349300,561380	3/3	32.5	73.4	5.7	N
27B	Odenhausen	349300,561380	3/3	36.5	72.9	6.3	N
28	Niederofleiden	359905,562365	3/3	179.0	−51.9	1.6	N
29	Neiderohmen	350235,561380	3/3	20.1	48.5	2.2	N
30	Oberohmen	350765,560915	3/3	350.3	52.0	5.0	N
31	Ulrichstein	351395,560420	3/3	196.2	−56.2	6.9	R
32	Schadges	353010,560325	3/4	180.2	−61.1	12.5	R
33	Rainrod	352470,561875	3/3	176.5	−72.6	9.0	R
34	Brauerschwend	352335,561875	–	–	–	–	–
35	Költenberg	352580,562160	3/3	249.5	35.2	36.9	I
36	Herbstein	352325,560165	3/3	355.5	63.7	7.3	N

Position refers to grid reference (R, H = easting, northing), n/N refers to the number of samples used for the mean over the number of samples stepwise demagnetized. D, Declination; I, inclination.

In the case of site 10, most of the cores have a very strong NRM in various, and abnormal, directions. This is almost certainly as a result of a lightning strike. Samples from site 21 have a two-component magnetization; the softer component is normal and may be a viscous magnetization in the direction of the present Earth's field, whereas the harder component seems to be northwesterly with a negative inclination. The exact direction of this component is not well defined; therefore a mean direction for this site was not calculated. Samples from site 35 proved to be rather unstable

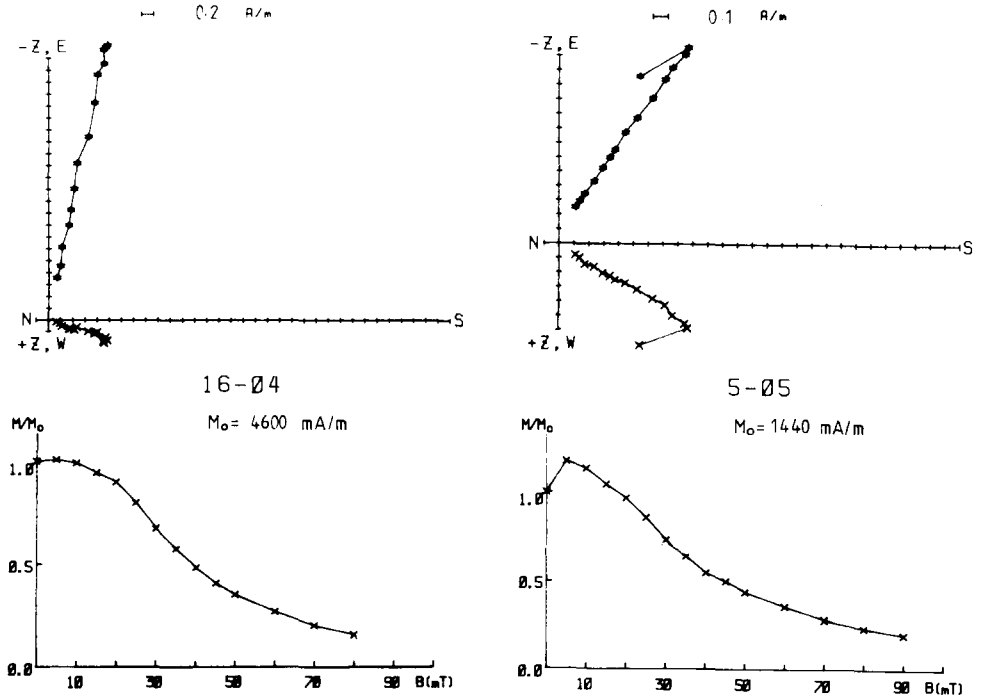


Fig. 2. Zijderveld (1967) and intensity plots of the stepwise AF demagnetization of two reversely magnetized samples ($\times$ = horizontal plane, $*$ = vertical plane).

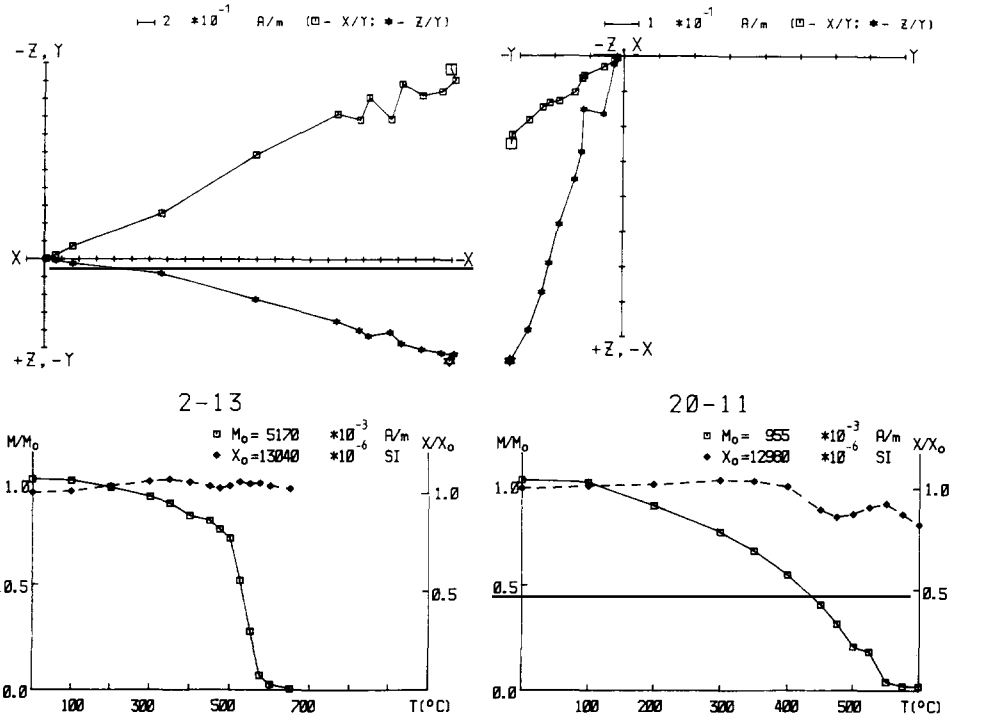


Fig. 3. Results of the thermal demagnetization of two unorientated samples, showing blocking temperature distribution and changes in susceptibility after each heating step.

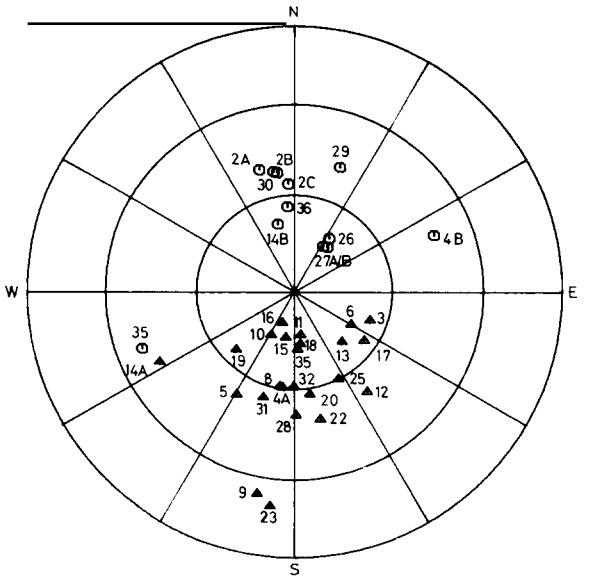


Fig. 4. Stereoplot of site mean directions (circles, positive inclination; triangles, negative inclination).

to AF demagnetization, so the errors are high and the intermediate (that is to say the pole lies $> 40°$ from the Earth's rotation axis) direction obtained may not reflect the geomagnetic field.

The site mean directions were calculated from the individual sample NRM directions using Fisher (1953) statistics (samples with clearly atypical directions were excluded at this stage). The mean magnetization directions are listed in Table 2 and plotted in Fig. 4. Out of the 36 sites with stable magnetizations, 10 have normal, 21 reversed and five intermediate directions.

In four places, samples were taken from two or more flows; in the cases of the three flows at Londorf (2A, 2B and 2C) and those at Odenhausen (27A and 27B), the flows all have similar directions; but at Vogelkopf (4A and 4B) and Grebenhain (14A and 14B) the two flows are of opposite polarity, in the former case reversed and nearly normal, and in the latter nearly reversed and normal.

4. Magnetostratigraphy

To define a reliable magnetostratigraphy, the relative ages of the sample sites must be known. In this case there are only sufficient stratigraphic and age data available to put the sample sites in an approximate order.

The magnetic time-scale of Cox (1982) shows that the following reversal sequence exists for the period at which the Vogelsberg is thought to have been most active (Fig. 5): normal 17.9–17.6 Ma (chron 5D); reversed 17.6–17.0 Ma (chron 5Cr); normal 17.0–16.2 Ma (chron 5C), with two reversed subchrons, 5C-1 (16.54–16.50 Ma) and 5C-2 (16.79–16.72 Ma); and reversed 16.20–15.23 Ma (chron 5Br). The ages quoted for these reversal boundaries have an associated error of about ± 0.2 Ma (Cox, 1982).

As part of some of the dating studies (Kreuzer et al., 1974; Harre et al., 1975; Ehrenberg et al., 1981), the polarities of samples have been measured (see Fig. 5). Within the error limits, the results are consistent with the magnetic time-scale (Cox, 1982). The oldest rocks, drilled in the central Vogelsberg, are mainly of normal polarity (Ehrenberg et al., 1981). Although there are a number of features which do not fit with the time-scale, from the age data it is reasonable to correlate these rocks with chron 5D, though they may in part be earlier. The rocks in the upper part of this hole are reversely magnetized and dated at ~ 17.6 Ma; these almost certainly formed during chron 5Cr, as were the rocks in the bottom half of the Rainrod I borehole (Kreuzer et al., 1974). The normally

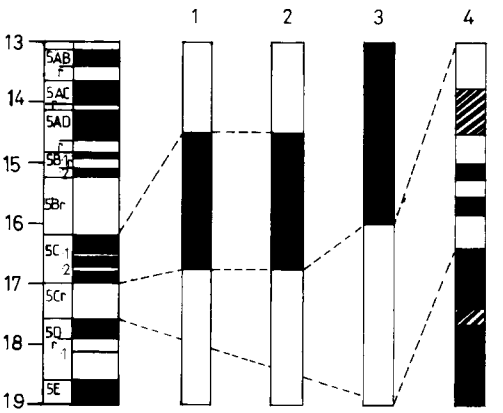


Fig. 5. Magnetostratigraphic correlations for the Vogelsberg using previously published data and the present results. Magnetic time-scale after Cox (1982). (1) Present study; (2) Harre et al. (1975); (3) Kreuzer et al. (1974); (4) Ehrenberg et al. (1981).

magnetized rocks in the top part of this hole may be correlated with chron 5C. Harre et al. (1975) stated that in the northwest Vogelsberg a reverse–normal–reverse (R–N–R) magnetostratigraphy exists, with the normal interval between 280 and 420 m above sea level. This assumes a simplified view of the actual stratigraphy which only applies to this small area. From the K–Ar ages it seems that the normal rocks correspond to chron 5C, and the upper reversed rocks to chron 5B. Whether the lower reversed rocks were formed during one of the reversed subchrons in chron 5C or during chron 5Cr cannot be determined from the available age data. In this area there are also a number of intrusive bodies which appear to be significantly younger than the main volcanic phase (Harre et al., 1975): these cannot easily be correlated with the magnetic time-scale.

A number of the sites sampled in this study have been previously dated by K–Ar (Harre et al., 1975; Ehrenberg et al., 1977). The ages quoted here have been corrected for the new values of the decay constants (Steiger and Jäger, 1977) to allow direct comparison with Cox's (1982) magnetic time-scale. The sites, with their suggested correlations, are as follows: Beuern (25), 16.75 ± 0.3 Ma, reversed, 5Cr or 5C-2; Londorf (26), 16.1 ± 0.3 Ma, normal, 5C; Vogelkopf lower flow (4B), 16.3 ± 0.3 Ma, intermediate (nearly normal), end chron 5C; and Vogelkopf upper flow (4A), 14.9 ± 0.5 Ma, reversed, ? chron 5Br.

Although there are no age data available from the other sites, it seems probable from the stratigraphic relationships that the rocks sampled in the southern central Vogelsberg (Fig. 1), which are reversely magnetized, were formed during the reversed chron 5Cr (17.6–17.0 Ma). The reversely magnetized rocks further from the centre probably formed more recently, perhaps during one of the two reverse subchrons during the 5C or during chron 5Br. Some of them may even be younger than 15.23 Ma (the end of chron 5Br (Cox, 1982)). All the normally magnetized rocks sampled probably formed during chron 5C (17.0–16.2 Ma). The proposed correlations between the paleomagnetic data from this and the earlier paleomagnetic studies and the magnetic time-scale are shown in Fig. 5.

5. Magnetic mineralogy and rock magnetism

To discover the effective grain-size distribution of the magnetic minerals, a number of rock magnetic parameters (Table 3) were measured.

Reflected light microscopy of polished sections shows that the magnetic mineralogy is simple, with the only contribution coming from titanomagnetite and its high-temperature oxidation products. In general, the degree of oxidation is low (see Table 3). Several specimens have no sign of deuteric oxidation (class 1; Haggerty, 1976), whereas the majority may be classified as 2 or 3 (Haggerty, 1976) and have 'exsolution' lamellae of ilmenite cross-cutting the titanomagnetite, which has moved towards magnetite in composition. In only one case was clear evidence of very high (class 7) deuteric oxidation seen, with pseudobrookite and titanohematite intergrowths. As this flow is vesicular in places and is only ~ 3 m thick, it seems likely that the oxidation followed the 'quick' path described by Haggerty (1976): class 1–class 2–class 7. Low-temperature oxidation was observed in a few samples, with the development of titanomaghemite generally occurring at grain boundaries and along cracks.

Measurement of the Curie temperatures of the samples using a horizontal magnetic translation balance, heating in air at $30\,^{\circ}C\ min^{-1}$ in an applied field of ~ 0.7 T, seems to confirm the optical observations. The results for the 41 sites (Fig. 6, Table 3) may be summarized as follows: 12 samples with low ($< 320\,^{\circ}C$) Curie temperatures, four with two Curie points (one high and one low), and 25 with high ($< 500\,^{\circ}C$) Curie temperatures.

Those with low Curie temperatures contain titanomagnetite unaffected by deuteric oxidation. Low-temperature oxidation to titanomaghemite leads to a rise in Curie temperature (O'Reilly, 1984), and as the average titanomagnetite in basalts has ~ 60% Ti (Haggerty, 1976), the Curie temperature of a fresh unoxidized sample is ~ $140\,^{\circ}C$. In the case of a sample from site 24 a maghemite disproportionation peak (O'Reilly, 1984) was observed, but in the other samples there was no incontrovertible evidence of major low-temperature oxidation, although many of the

TABLE 3

Rock magnetic data

Site	M_{NRM} (mA m^{-1})	M_{10} (mA m^{-1})	K ($\times 10^{-6}$)	K_{FD}%	Q	M_{RS} (A m^{-1})	$(B_0)_{CR}$ (mT)	T_c (°C)	Oxidation class	L T K group	K_{78}/K_{293}
1	3160	3222	28560	2.6	2.96	1318	53	510	2	3	1.26
2A	26470	25300	27500	0.8	25.7	975	47	585	7	3/2	1.09
2B	2183	2087	4100	6.2	14.2	78 [a]	48	620	7	1	0.95
2C	1480	1047	10637	1.3	3.84	319	17	190/520	1	3/1	0.50
3	2253	1542	29680	2.5	2.16	428	12	160	1	3/1	0.55
4A	2327	1573	43183	7.5	1.44	145	12	65	1	1 [b]	0.10
4B	1797	830	45733	6.2	1.05	174	18	120	1	1	0.13
5	2546	2413	32720	1.1	2.23	686	37	540	2	3/1	0.40
6	1479	1208	46100	3.9	0.87	840	29	535	3	1	0.42
7	2420	2240	64300	6.5	1.01	239	15	130	1	1	0.09
8	6177	2957	12083	5.1	13.9	176	19	120	1	1	0.11
9	1540	1249	42467	3.1	0.97	632	34	515	2	2	1.30
10	2327	1286	31003	3.2	1.93	646	25	505	2	1	0.26
11	2140	2347	44033	1.4	1.38	1111	46	530	2	3/1	0.48
12	3650	3680	43117	0.6	2.26	1881	47	555	3	2	1.15
13	1085	681	27450	4.5	1.07	329	15	170/480	1 [c]	1	0.38
14A	769	772	16593	1.0	1.29	1193 [a]	75	580	7	2	1.55
14B	3563	3267	35400	0.6	2.70	2263	77	550	3	3	1.05
15	961	605	20073	0.3	1.30	192	16	540	2	2	2.35
16	5480	5227	16140	0.7	9.70	390	33	560	3	3	1.10
17	1460	1530	39000	0.3	0.99	1201	45	550	2	3	0.90
18	371	498	27583	4.7	0.34	197	11	85/475	1	1	0.50
19	412	295	12203	2.6	0.91	47	15	520	2	1	0.45
20	1777	1677	17837	0.1	2.67	475	42	565	2	1	0.70
21	325	110	17880	0.6	0.49	203	37	530	3	1	0.62
22	1608	1451	38343	2.5	1.49	638	54	510	3	1	0.15
23	1353	944	19527	3.2	1.84	414	12	210	1	1	0.32
24	854	458	4610	12.4	4.95	30	12	230	1	1 [b]	0.18
25	1767	1803	4493	3.2	10.5	681	37	550	3	1	0.25
26	2023	1673	2823	3.6	19.4	71	36	570	3	2	1.65
27A	3103	3020	4501	2.1	18.4	178	71	580	3	2	2.20
27B	3817	3390	7797	2.6	13.2	202	48	565	3	2	1.66
28	450	276	8653	2.8	1.37	49	12	540	2	2/1	0.85
29	2177	1403	12840	2.4	4.56	325	23	230/575	2	1	0.35
30	2737	2327	25540	0.6	2.90	509	46	565	3	1	0.18
31	1187	696	44800	3.4	0.70	357	13	300	1 [c]	1	0.18
32	2034	1333	13673	1.1	3.68	512	18	225	1	1	0.12
33	3380	2073	25673	2.3	4.25	321	12	100	1	1	0.08
34	856	506	5870	8.1	3.90	146	32	90	1	1 [b]	0.40
35	806	214	40283	2.9	0.53	440	17	220	1	1	0.13
36	9973	6764	25567	4.1	10.4	789	48	580	3	1	0.30

[a] Samples did not saturate in a field of 1.6 T.

[b] Samples had a Hopkinson peak (4A×1.2 at ~15°C, 24×1.6 at ~5°C, 34×2.9 at −20°C).

[c] Low-temperature oxidation was clearly visible in samples.

curves were irreversible, which suggests that a small amount of low-temperature oxidation had occurred in the field. The samples with two Curie points have been deuterically oxidized, but unoxidized titanomagnetite remains in some places. Most of the samples are deuterically oxidized; the Ti has gone into ilmenite lamellae, so that the remaining spinel phase approaches magnetite in

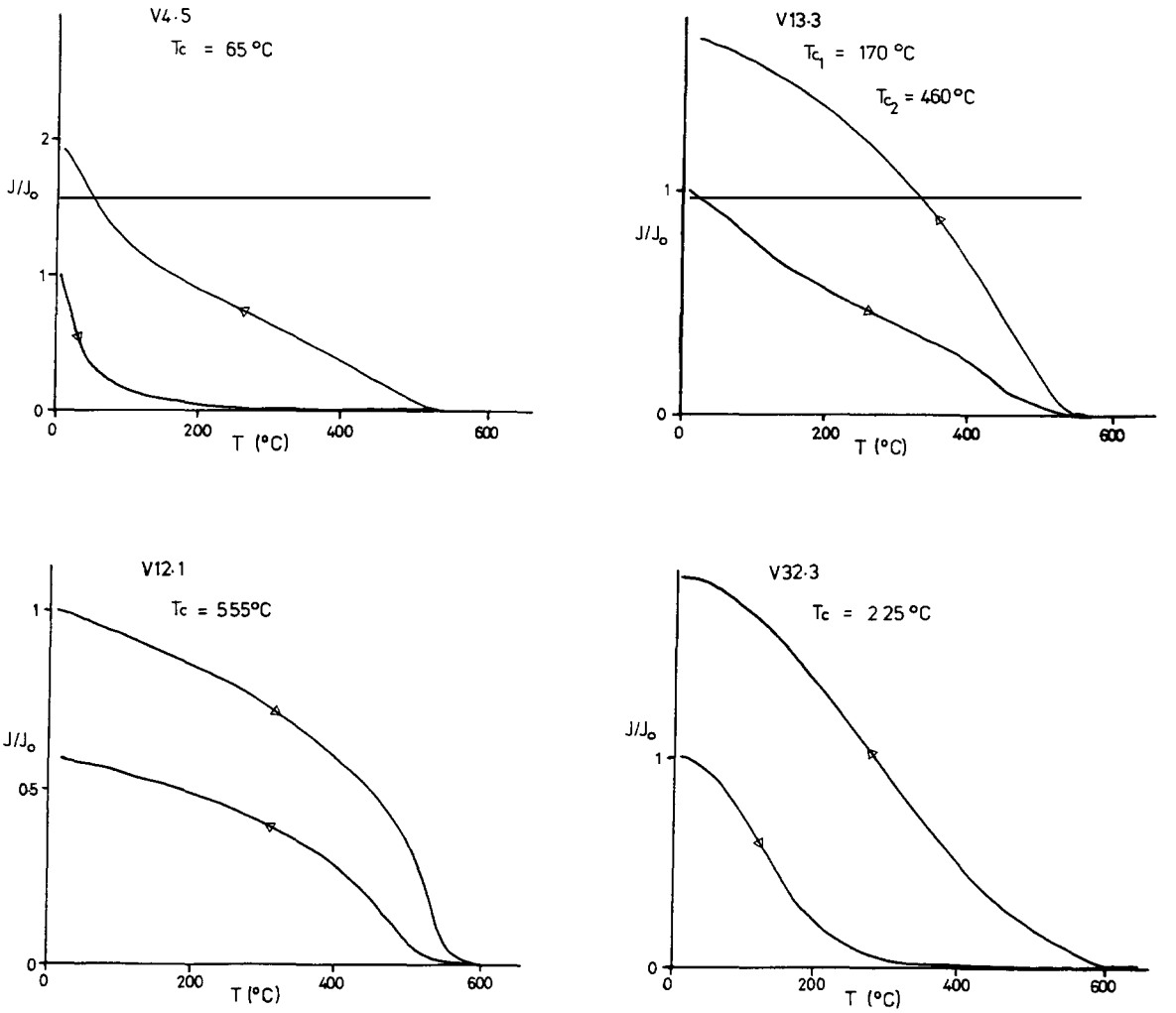


Fig. 6. Examples of thermomagnetic behaviour: samples displayed a wide range of Curie temperatures and degrees of thermal alteration.

composition, hence the high Curie temperature. The sample from site 2B, which optically seems to be class 7, has a Curie temperature of 620 °C, which is consistent with a iron-rich titanohematite.

There seems to be a relationship between the amount of deuteric oxidation of the titanomagnetites and the form of the rock. All samples from vesicular flows, which tend to be only a couple of metres thick and were rich in volatiles, have oxidized titanomagnetites, whereas the fine-grained massive rocks contain often completely unoxidized titanomagnetites.

The NRM intensity (M_{NRM}) and the intensity of the NRM remaining after AF demagnetization to 10 mT (M_{10}) are listed (Table 3). Bulk susceptibility (K) was measured using a KLY-2 kappa-bridge, and its frequency dependence (K_{FD}%; Thompson and Oldfield, 1986) using a Bartington MS2 susceptibility bridge. The variation of the bulk susceptibility between −196 °C and room temperature was determined using a water-jacketed probe attached to the MS2 meter. From M_{NRM} and K, the Königsburger (1938) ratio (Q) has been calculated. Using an electromagnet with

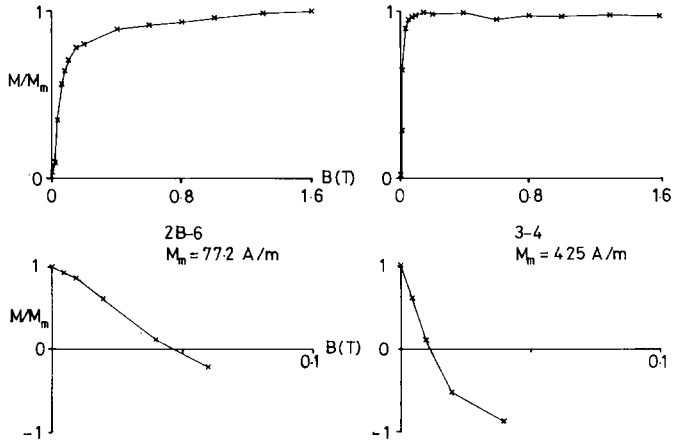


Fig. 7. Examples of IRM acquisition and backfield IRM curves. Sample 2B-6 does not saturate at 1.6 T, suggesting the presence of ilmenohematite in addition to magnetite. Sample 3-4 saturates at relatively low fields, indicating that titanomagnetite is the magnetic carrier.

a maximum field of 1.6 T, isothermal remanent magnetization (IRM) acquisition and back-IRM curves (Fig. 7) were determined. The values of the intensity of saturation remanence (M_{SIRM}) and the coercivity of remanence ($(B_0)_{CR}$) have been determined.

The values of the susceptibility, NRM and saturation IRM give an indication of the amounts of magnetic minerals present. For grain-size distribution the Q ratio, $K_{FD}\%$, and $(B_0)_{CR}$ are the most important indicators. Q is supposed to be ~ 0.5 for a multidomain (MD) grain (McElhinny, 1973), so that samples with high Q values have mainly single domain (SD) grains. $K_{FD}\%$ for a MD grain is virtually zero and the largest effects are seen with grains on the SD–superparamagnetic boundary (Thompson and Oldfield, 1986). The value of $(B_0)_{CR}$ is related to the size and shape of the grains as well as their composition. For hematite it is usually > 100 mT, whereas for SD magnetite with elongated grains it is between 40 and 75 mT, and for MD titanomagnetite ($X = 0.6$) it is between 10 and 20 mT (Thompson and Oldfield, 1986). The low-temperature susceptibility behaviour can be split into three groups (Senanayake and McElhinny, 1981): group 1, where the susceptibility increases with temperature; group 2, where there is a decrease in susceptibility with increasing temperature; and group 3, where there is a peak in susceptibility at around $-155\,^{\circ}$C. The exact interpretation of the three groups is open to debate (Sherwood, 1988a), but it is generally agreed that Ti-rich titanomagnetites exhibit group 1 behaviour, group 2 behaviour is restricted to deuterically oxidized samples and seems to be caused by stress effects in SD Ti-poor magnetite grains which have formed between the exsolved ilmenite lamellae, and the peak observed in group 3 samples marks the Verwey transition in MD magnetite.

In general, deuteric oxidation and the production of ilmenite lamellae will cause subdivision of the grains and result in a decrease of effective grain size. Whether the Ti-poor titanomagnetite that is produced is MD or SD will depend on the coarseness of the ilmenite lamellae. The actual samples (Table 3) are not so easy to categorize, often certain parameters will suggest SD assemblages, whereas others suggest MD grains. This can be explained to some extent by inhomogeneities in the degree of oxidation and the grain size distributions and also by mixtures of grain sizes and compositions of magnetic minerals.

In three cases the median destructive field is < 10 mT, and in five cases, because of the removal of a viscous component, the intensity is higher after the AF demagnetization. In two instances, sites 2B and 14A, it proved impossible to saturate the samples in a field of 1.6 T. This suggests the presence of goethite or titanohematite

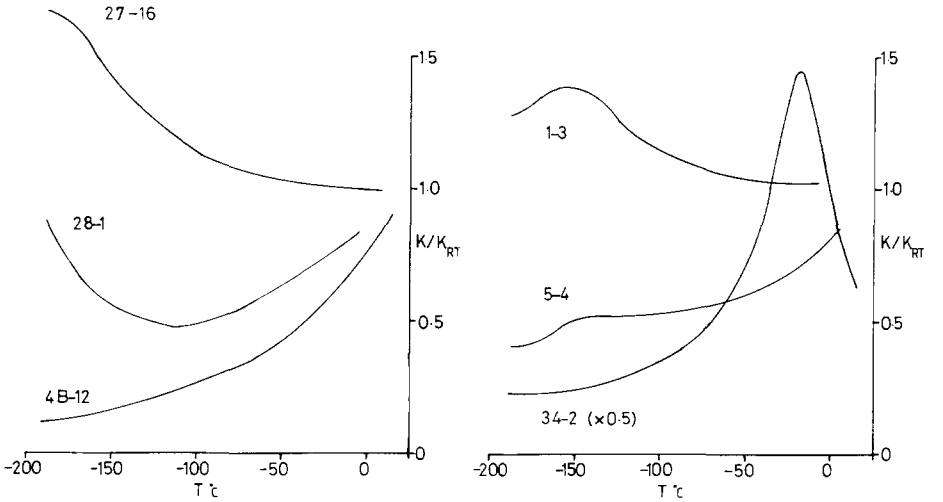


Fig. 8. Examples of low-temperature susceptibility behaviour. The samples are classed as follows: 27-16, group 2; 28-1, group 2/1; 4B-12, group 1; 1-3, group 3; 5-4, group 3/1; and 34-2, group 1 with Hopkinson peak.

in these rocks. From thermal demagnetization studies it is clear that this unsaturated phase is titanohematite rather than goethite.

Six samples with low Curie temperatures have strongly frequency-dependent susceptibilities: of these, three (sites 4A, 24 and 34) have group 1 low-temperature susceptibility behaviour with a peak just below room temperature (Fig. 8) — this Hopkinson peak is due to grains exceeding their blocking temperature and becoming superparamagnetic.

The low-temperature susceptibility behaviour of synthetic titanomagnetites and magnetites was described by Senanayake and McElhinny (1981). Their results show that group 1 behaviour is found in synthetic titanomagnetites of most compositions but the ratio K_{78}/K_{293} increases with decreasing Ti content from < 0.2 for $X = 0.6$ to 0.9 for SD magnetite. This can be seen in natural samples which show group 1 behaviour (Table 3, and Sherwood, 1988a), with a correlation between the ratio of susceptibility at liquid nitrogen temperature to room temperature and the Curie temperature. The peak due to the Verwey transition (group 3) was not seen in the Ti-rich ($X = 0.3$ and 0.6) titanomagnetites of all grain sizes, whereas it is present in MD Ti-poor titanomagnetites. This suggests that the samples which show a group 3 type peak superimposed on group 1 type curves (group 3/1; Fig. 8 and Table 3) contain a mixture of Ti-rich titanomagnetite and MD Ti-poor titanomagnetite; this is supported in the case of flow 2C by the existence of two Curie temperatures. The sample from site 28 has an interesting low-temperature susceptibility curve (Fig. 8); the susceptibility initially decreases as the rock warms from liquid nitrogen temperature and then at about $-100\,^{\circ}$C begins to increase. This behaviour is classified as group 2/1. Optical analysis of this sample reveals primary ilmenite and titanomagnetite with some deuteric oxidation. From the theoretical calculations of Senanayake and McElhinny (1981), it is possible to obtain this saucer-shaped form of the stress-induced group 2 behaviour with certain compositions and grain shapes.

6. Mean VGP and the European APWP

The mean virtual geomagnetic pole (VGP) for the Vogelsberg samples, excluding those with intermediate directions (with VGP latitudes $< 50\,^{\circ}$) is 85.1° N, 200.9° E. The previous results from the Vogelsberg (Angenheister, 1956; Nairn, 1960), together with other VGPs from central European volcanic centres active at about the same time as

TABLE 4

VGPs for central European volcanics (20–10 Ma) (data from Weinreich and Bleil, 1984)

Area	Age (Ma)	D	I	α_{95}	P_{lat}	P_{long}
Kaiserstuhl	18–13	5.7	50.1	12.6	72.2	171.3
Ries Impact Crater	15	11.4	59.4	1.4	78.4	142.6
Göttingen area	20–13–7	11.7	61.2	8.9	77.9	143.9
Habictswald	Late Miocene	357.7	66.2	19.1	86.9	218.9
Hegau	16–13	8.3	65.0	10.5	84.2	111.4
NW Vogelsberg (Angenheister, 1956)	18–15	13.9	57.4	5.1	76.0	160.0
Vogelsberg (Nairn, 1960)	18–15	–	–	–	86.0	168.0
Vogelsberg (this study)	18–15	358.6	64.0	5.2	85.1	200.9

the Vogelsberg (Weinreich and Bleil, 1984), are given in Table 4.

The previous two mean poles from the Vogelsberg were obtained using only NRM data, although from the present study it appears that in most cases a viscous component does not significantly contaminate the original magnetization direction. Angenheister (1956) collected samples from over 40 flows situated in the northwestern Vogelsberg. Many of these sites are no longer exposed, but repeat measurements of some of his sites (this study — 26, 27A, 27B and 29) confirm that in this part of the Vogelsberg several flows, which presumably were erupted during a very

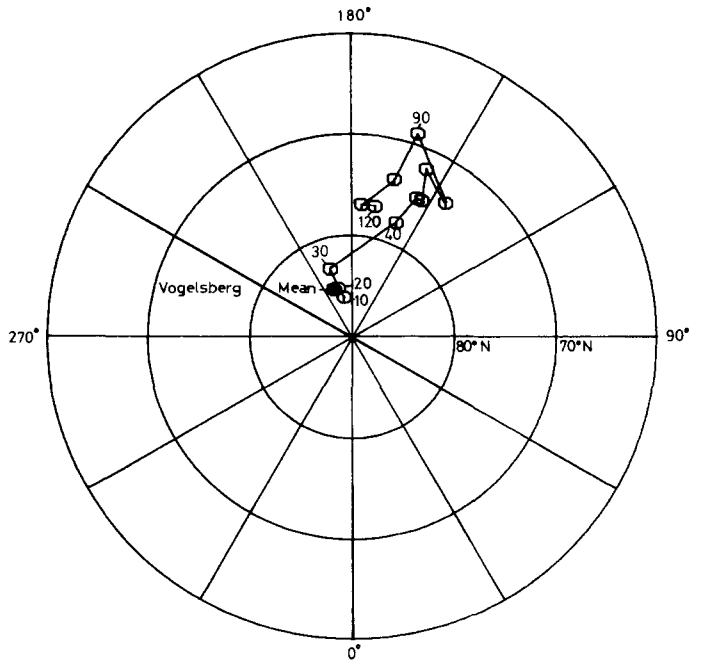


Fig. 9. The Vogelsberg mean VGP plotted on the European APWP (Irving and Irving, 1982).

short period of time, have a remanence direction with a declination of $\sim 30°$. This would explain the easterly mean direction obtained by Angenheister. Nairn's (1960) pole is not dissimilar to the present one. However, he provided no details of sampling localities.

The mean VGPs from other central European volcanics (Table 4) are not very similar to the pole obtained in this study. However, in no case did the number of sample sites exceed 15 (Weinreich and Bleil, 1984), so good averaging of secular variation is unlikely to have been achieved. The Ries impact crater is a special case in which the result obtained represents a single 'snapshot' of the geomagnetic field at the time of the meteorite impact (hence the very small α_{95}).

A number of the poles used by Weinreich and Bleil (1984) and listed in Table 4 were also used by Irving and Irving (1982) in their calculation of a Eurasian apparent polar wander path (APWP). However, although the references used for the paleomagnetic data are the same, there are differences in the assigned ages of the rocks and in the positions of the poles. These discrepancies can be explained by the fact that Weinreich and Bleil (1984) recalculated their poles from the original data and rejected some results that had originally been included, and that they also used radiometric age determinations that had been published after the original paleomagnetic studies. The effect of these dating errors on the APWP is probably not too serious, as Irving and Irving used a 20-Ma smoothing window (20 Ma), and a large number of poles are averaged.

The APWP of Irving and Irving (1982) is plotted in Fig. 9, with the mean VGP for the Vogelsberg volcanics. The new result falls almost exactly on the portion of the path between 10 and 20 Ma, which is consistent with the age of the rocks.

Weinreich and Bleil (1984) have demonstrated that there is a marked difference between Irving and Irving's Late Tertiary APWP, which travels up the 180° meridian, and one calculated from the central European volcanics alone. Work on Miocene volcanics in New Zealand (Sherwood, 1988b) has shown that it is possible for paleomagnetic poles obtained from volcanic centres to be a poor representation of the time-averaged geomagnetic field even if a large number of sites are used.

The new data from the Vogelsberg appear to support the suggestion of Weinreich and Bleil (1984) that the difference between the two paths is caused by inadequate averaging of the geomagnetic field by the samples from the other volcanics, but more data from stable Europe for the Miocene are still required to give a reliable APWP that can be used when comparing results in the tectonically deformed parts of Eurasia.

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